

Atharva Kulkarni

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Offensive Security · Penetration Testing | linkedin.com/in/atharva1/ | atharvaxsecurity.com

Summary

CEH V12 certified with a **GCHQ-accredited MSc in Applied Cyber Security**, moving into offensive security and penetration testing. Three years of enterprise IT and infrastructure experience across **Windows**, **Active Directory**, endpoint compliance and networking, now applied to **security assessment of production systems** — identifying vulnerabilities, tracing root causes and driving fixes to closure with the teams who own them. Builds and ships **security tooling**: a 50+ tool client-side toolkit with 240 automated tests, and a multi-agent automation framework. **Top 2% on TryHackMe** across 131 completed rooms, with an offensive MSc dissertation applying **generative adversarial networks** to hardware security primitives. Working knowledge of **OWASP Top 10**, **MITRE ATT&CK** and the **Cyber Kill Chain**. Targeting penetration testing and red team operations.

Experience

Technical Support Analyst | Eurostop, United Kingdom

09/2025 – Present

- Conduct ongoing **security assessments** of internal systems including a cloud-based POS platform, stock control system, internal integrations and a locally hosted **LLM service**, identifying vulnerabilities and contributing to the organisation's security posture.
- Deliver end-to-end support for enterprise **EPOS platforms**, overseeing transactional and operational data flow across in-store systems, **middleware**, head-office platforms and **third-party integrations**.
- Diagnose and resolve data consistency, pricing and product-lifecycle issues through cross-system analysis, identifying **root causes**, discrepancies and process failures within large data sets.
- Support external system integrations, **incident investigation** and operational controls, ensuring **secure access**, reliable synchronisation, data integrity and continuity across retail environments.

DSS Engineer | Diageo, United Kingdom

02/2025 – 09/2025

- Implemented and advised on **security-focused endpoint management**, including **Intune-Autopilot** integration, tenant lock and device compliance enforcement, strengthening asset protection and reducing exposure.
- Resolved issues across **Active Directory**, authentication, Wi-Fi and device compliance; contributed to **Zero Trust**-aligned access controls and centralised asset tracking via **Nexthink** and **ServiceNow**.
- Delivered frontline IT support for enterprise users across hardware, network, **Windows OS** and application-level issues, focused on **asset security and user account management**.

Technical Support Engineer | Concentrix, United Kingdom

01/2024 – 01/2025

- Advised customers on **cybersecurity scams and phishing techniques**, reducing security breaches and raising cyber awareness across a broad customer base.
- Provided technical support to BT Broadband customers, improving query resolution efficiency by **35%**.
- Troubleshoot broadband services, mobile telephony and home tech products, producing an **80% decrease** in escalated tickets.

Network Operations Specialist | Clariant India Ltd, Mumbai

01/2022 – 07/2022

- Enhanced organisational security through **endpoint security configurations**, contributing to a **25% decrease** in security breaches and a 20% reduction in security incidents.
- Maintained and configured **network switches and routers**; developed **cybersecurity awareness training**, increasing employee security awareness by 15%.
- Provided technical support via **ServiceNow**, reducing network outage response time by **30%** and configuring over 250 machines.

Security Tooling & Automation

Client-Side Security Toolkit — atharvaxsecurity.com/cybersec-toolkit

- Built and maintain a browser-based toolkit of **50+ self-authored security tools**: encoding and cipher analysis, hash identification and cracking, **JWT / AES / RSA** and **X.509** inspection, **CVSS 3.1** scoring, secret scanning, steganography, homoglyph detection and **SPF / DKIM / DMARC / BIMi** email-authentication analysers.
- **240 automated tests**, verified against **RFC 4648**, **FIPS 180-4** and **RFC 4231** published vectors. Integrated the **CyberChef** engine (505 operations) to run fully client-side under a custom interface.

AI Agent Orchestration Framework — github.com/atharvak161/claude-org-framework

- Designed and maintain a multi-agent operating framework of **89 role-scoped agents** across 12 departments, with enforced review gates, **security sign-off before release** and structured audit logging.

- Authored and published a **248-note** reference base covering cloud security, **GRC**, **OSCP** preparation and red-team technique, with full-text search and cross-linking.

Education

MSc Applied Cyber Security | Queen's University Belfast 2022 – 2023

GCHQ-accredited. Modules: **Network Security & Monitoring**, **Malware Analysis**, Software Assurance, **Computer Forensics**, Applied Cryptography. Dissertation: **Machine Learning Attacks on Physical Unclonable Functions**.

BSc Computer Science | Pillai's College of Arts, Commerce and Sciences 2019 – 2022

Core modules in network technologies, routing protocols, **Cisco network infrastructure** and network security.

Skills

Offensive Security: Penetration Testing, Vulnerability Assessment, Network & Host Enumeration, Manual Exploit Identification, **Red Team Concepts**, CTF & Lab Practice

Web & Application: OWASP Top 10, Burp Suite, Web Application Security Testing, JWT / AES / RSA, **X.509** Certificate Analysis

Threat & Analysis: MITRE ATT&CK, Cyber Kill Chain, OSINT, Indicator Extraction, **CVSS 3.1** Scoring, Secret Scanning, Entropy Analysis

Infrastructure: Active Directory, Intune & Endpoint Compliance, **Zero Trust** Access Controls, Firewall Configuration, TCP/IP, Network Troubleshooting

Engineering: Python, Bash, JavaScript, Git, Automated Testing, SQL, Security Tooling Development

Platforms & Tools: Kali Linux, Nmap, Metasploit, Wireshark, ServiceNow, Nexthink, TryHackMe, HackTheBox

Certifications

- **Certified Ethical Hacker (CEH V12)** — EC-Council, ECC9421760853
- **Cyber Security 101 (SEC1)** — TryHackMe, 2026
- **Pre Security (SEC0)** — TryHackMe
- **Fortinet NSE Level 1** — PggZRhVh2p · **Fortinet NSE Level 2** — uTMYfCWHCd
- **TryHackMe Junior Penetration Testing Path** — in progress, **top 2% globally**, 131 rooms completed

Projects

Machine Learning Attacks on Physical Unclonable Functions | MSc Dissertation 2023

Investigated machine learning vulnerabilities in hardware security primitives. Employed **generative adversarial networks** to synthesise PUF challenge-response pairs, enabling **modelling attacks** and evaluation of countermeasures. Offensive security research at graduate level.

Honey Encryption: Enhancing Password-Based Security | Queen's University Belfast 2023

Investigated **Honey Encryption** as a defence against **brute-force attacks** on password-protected systems. Designed a framework and developer API generating realistic decoy plaintext on incorrect password attempts.

Digital Forensics: NTFS File Recovery | Queen's University Belfast 2022

Forensic investigation of deleted file recovery on Windows 10 **NTFS** using command-line tools, **disk imaging**, **MFT analysis** and slack space examination. Applicable to **incident response** and data breach investigation.